

Title: Model Boxer Engine Analysis

Section 1 Group 1

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BACKGROUND AND OBJECTIVE

- Design a crankshaft mechanism modeled after the Subaru EJ25
- Selected model for its over-square design (bore > stroke)
- Four-cylinder boxer with bore and stroke scaled down by a factor of 5
- Analyze kinematics of the piston mechanism
- Objective #1: sustain 900 RPM for the duration of a minute
- Objective #2: sustain a higher RPM than our peers

PROJECT CONSTRAINTS AND REQUIREMENTS

- Must fit within a scaled-down 2012 F-150 engine bay
- Cannot be an eight-cylinder system
- Fall within: 131 mm x 131 mm x 102mm and 80 mm x 80 mm x 80 mm
- Compression Ratio < 14:1
- Must safely mount and fit into the designated tester
- Must operate smoothly without the use of lubricants

DESIGN PROCESS AND ITERATIONS

- 4-cylinder boxer
- Crankshaft made from separate pieces
- Biggest jump was from prototype 1 to prototype 2
- First Iteration
 - Multiple failing points
 - Cylinder, fitment, crankshaft, etc.
- Second Iteration
 - Crankshaft fusion
 - Didn't mount properly
- Final Iteration
 - Still experienced fusion

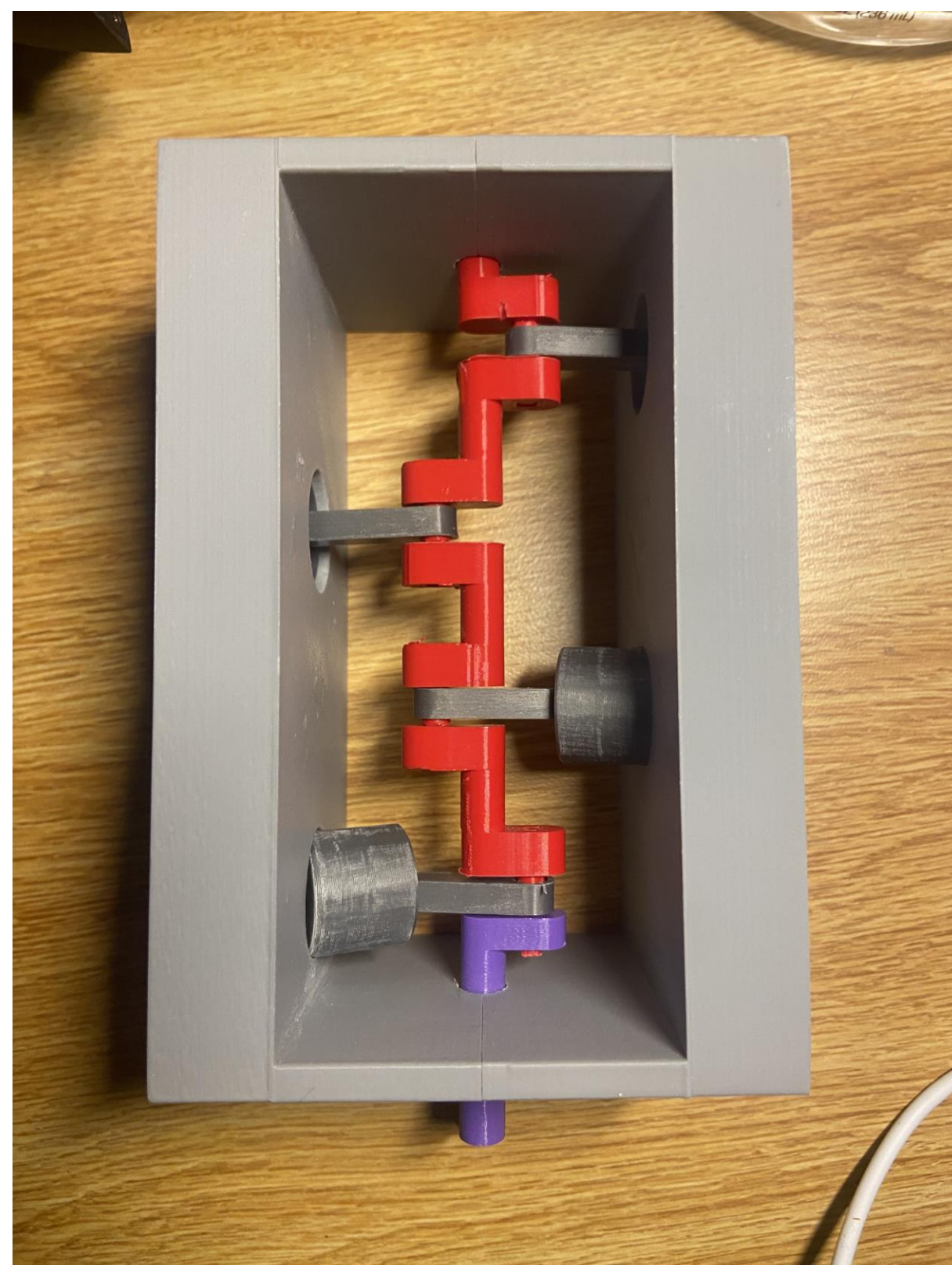


Fig. 1. Top view of first iteration.

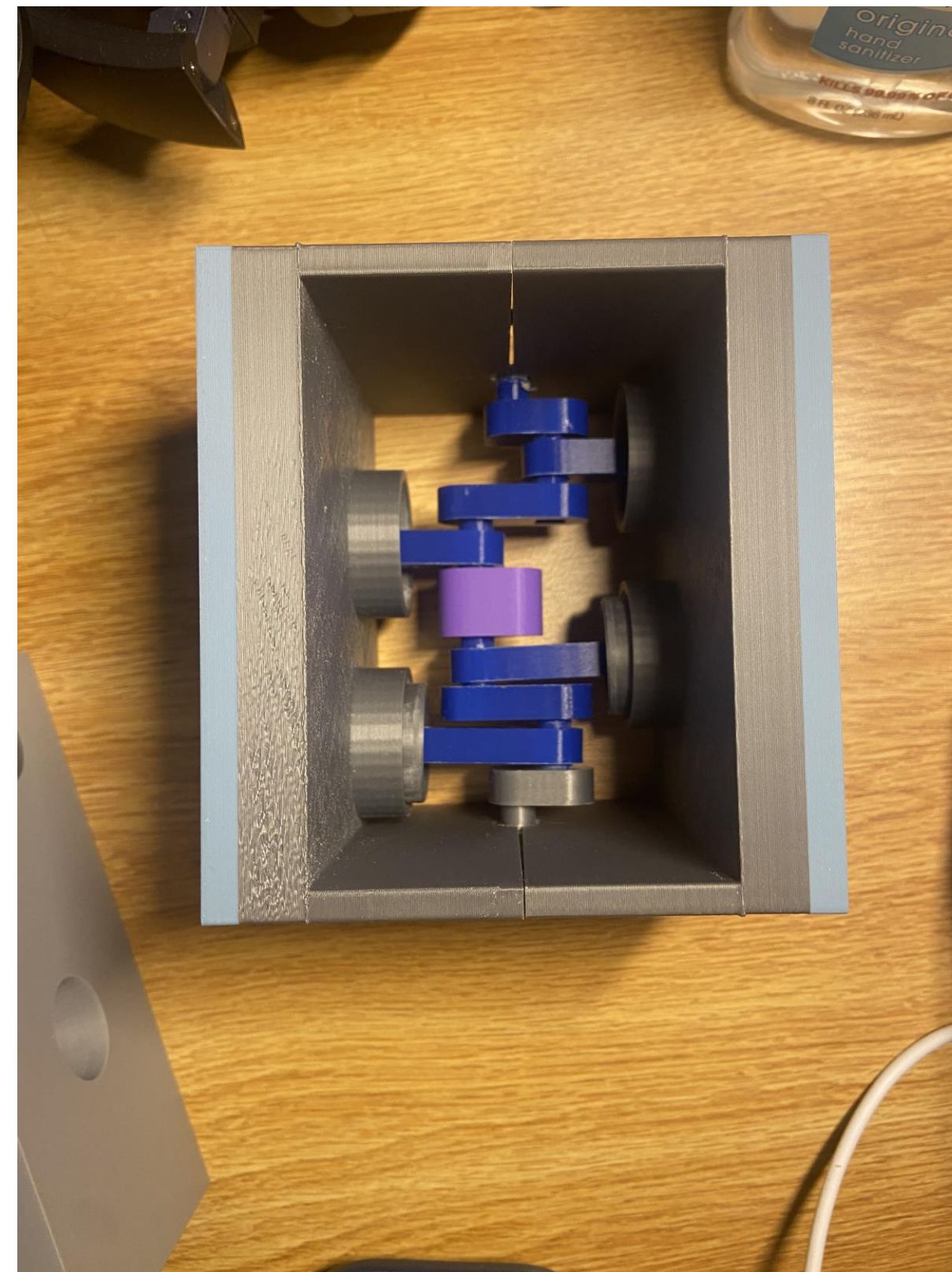


Fig. 2. Top view of second iteration (post-test).

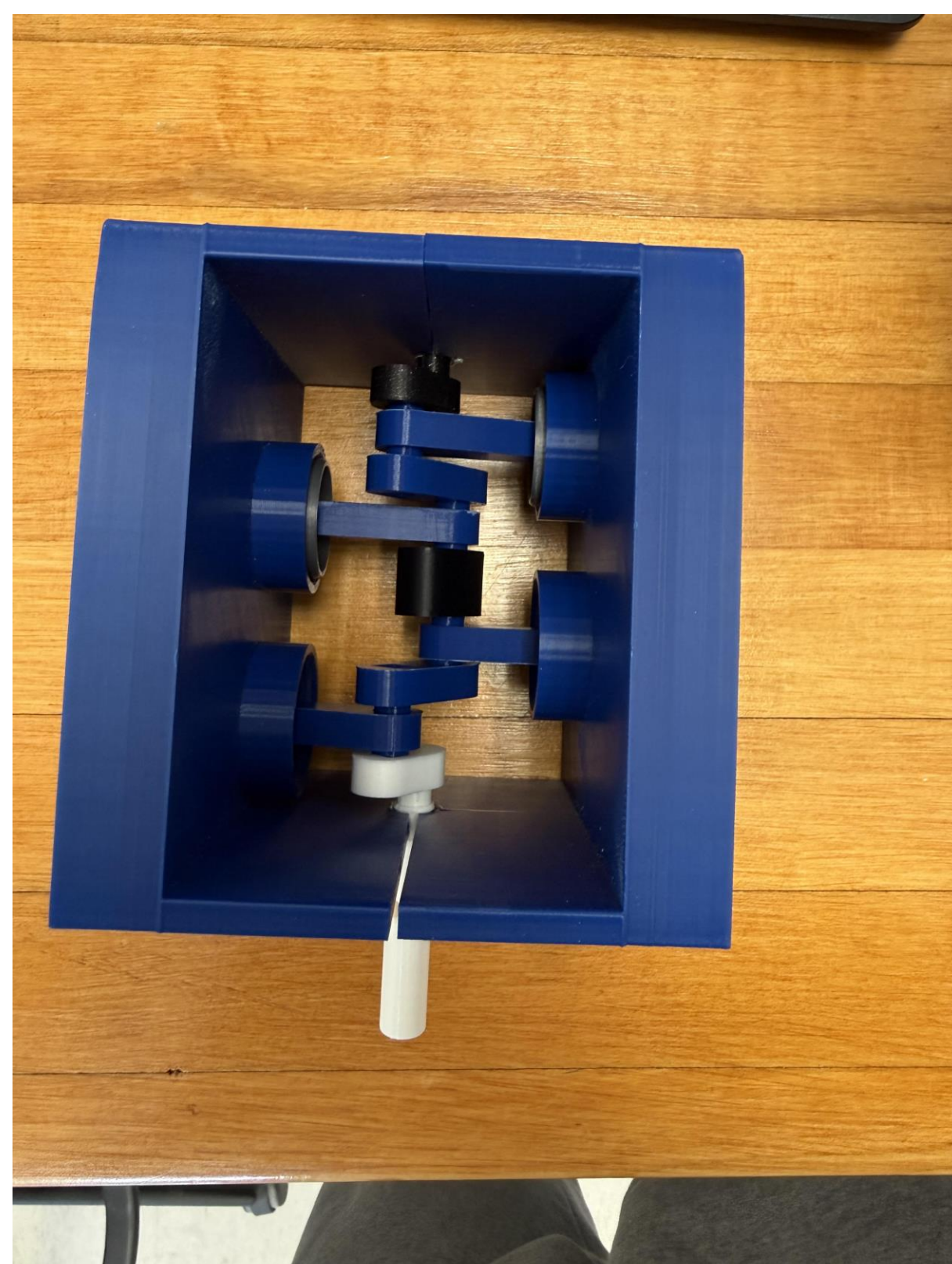


Fig. 3. Top view of final iteration (before test).

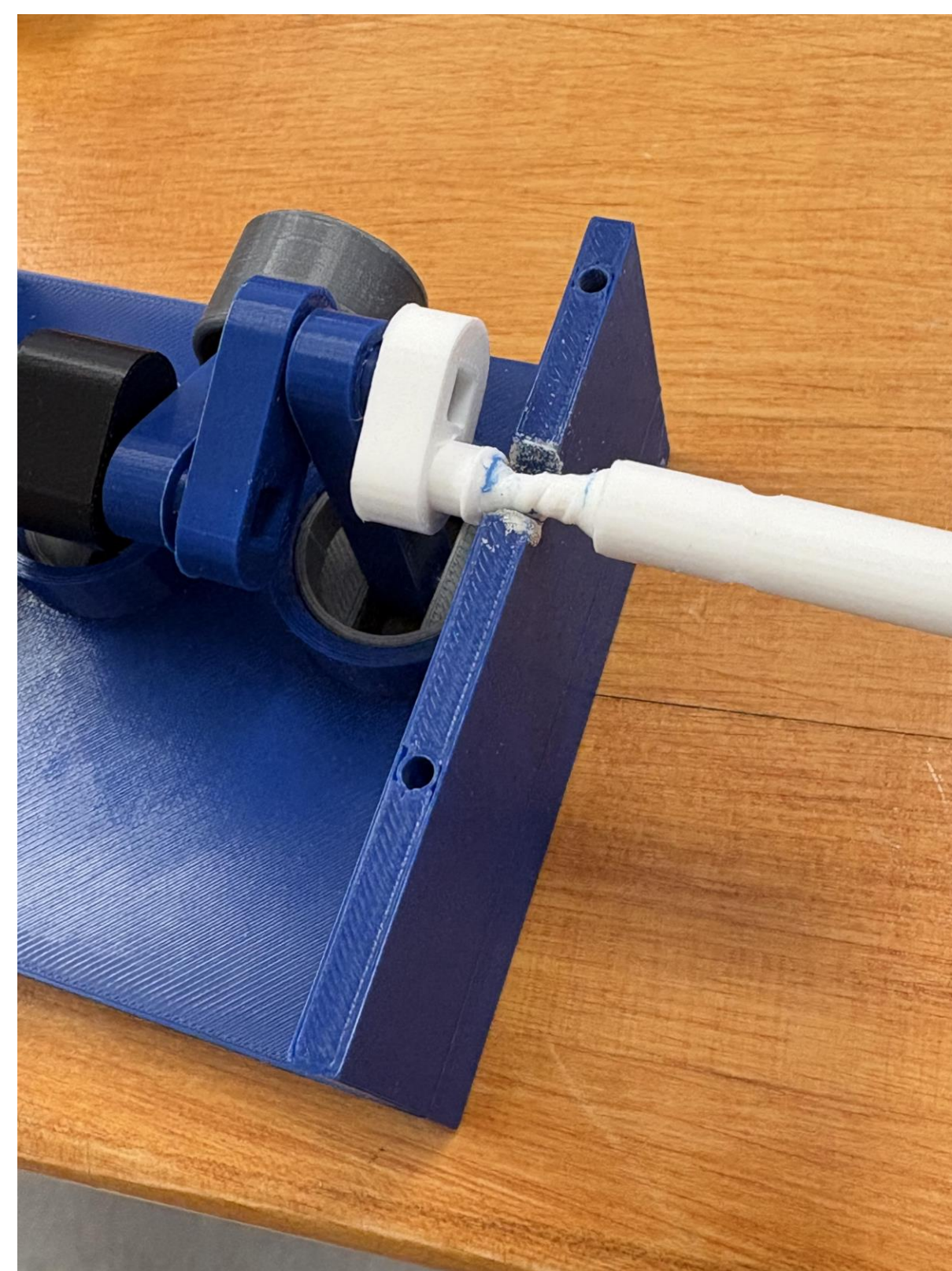


Fig. 4. Failure of crankshaft (after test).

RESULTS AND DISCUSSION

- Reached a maximum of 2100 RPM
- Velocity, acceleration, and stress were analyzed at 900 RPM.
- Average velocity was similar across theoretical, simulation, and experimental analysis.
- Average acceleration and average stress found in the SolidWorks deviated from theoretical values.
- Engine did not fail due to failure of wrist pin, rather due to fusion of a connecting rod to the crankshaft.

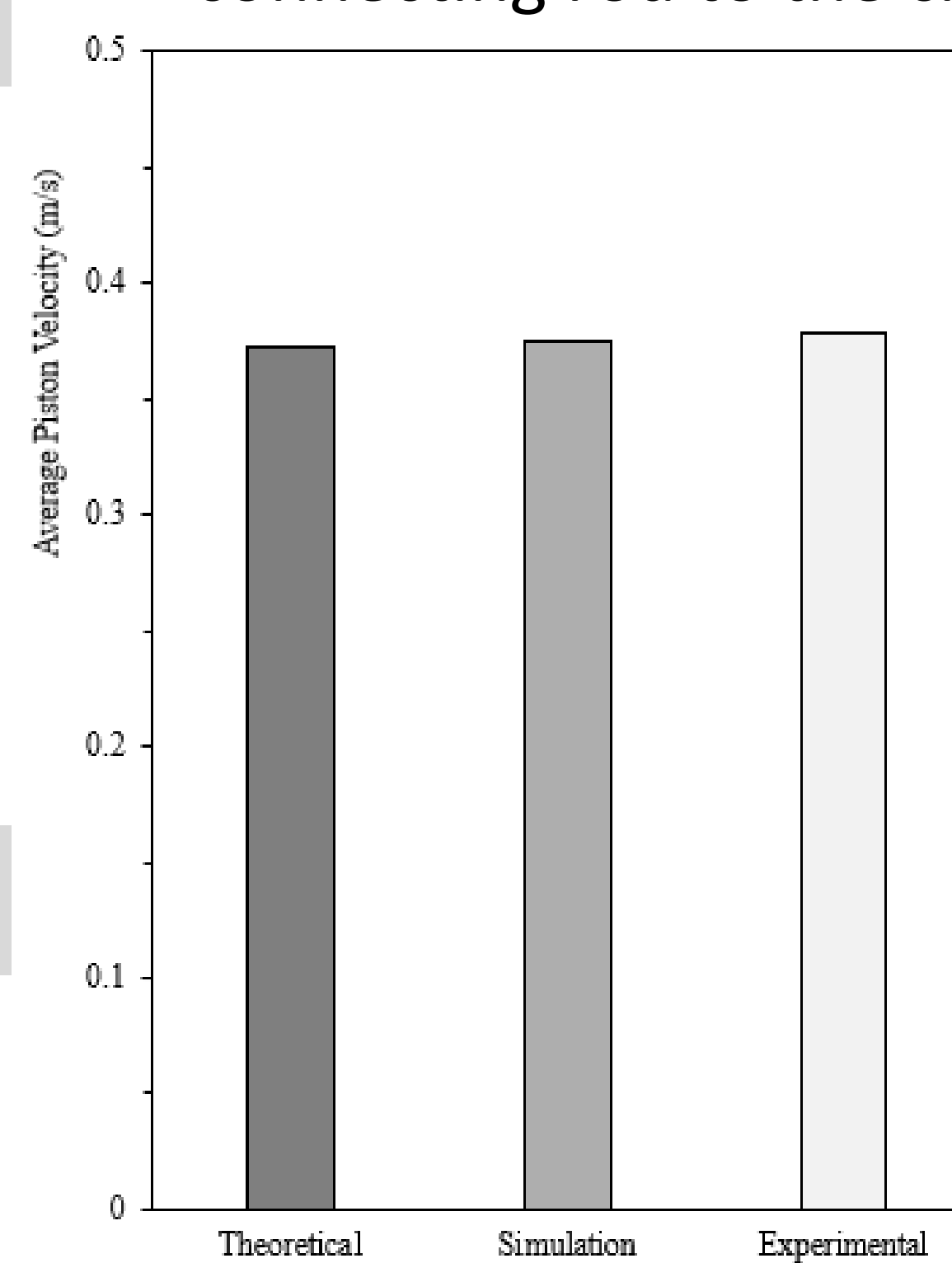


Fig. 5. Average Piston Velocity comparison between theoretical hand calculations, SolidWorks simulation, and experimental observation.

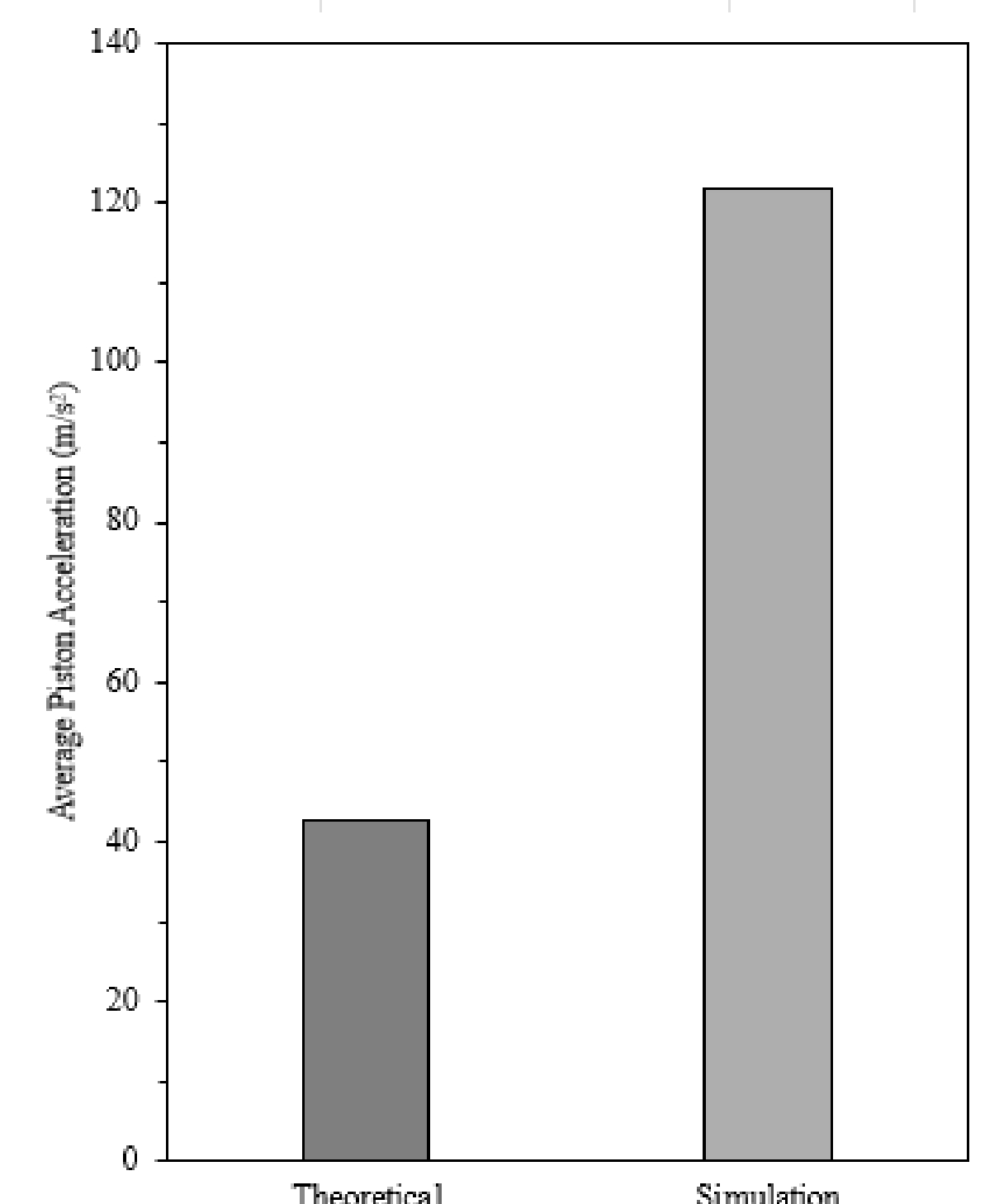


Fig. 6. Average Piston Acceleration comparison between theoretical hand calculations and SolidWorks simulation.

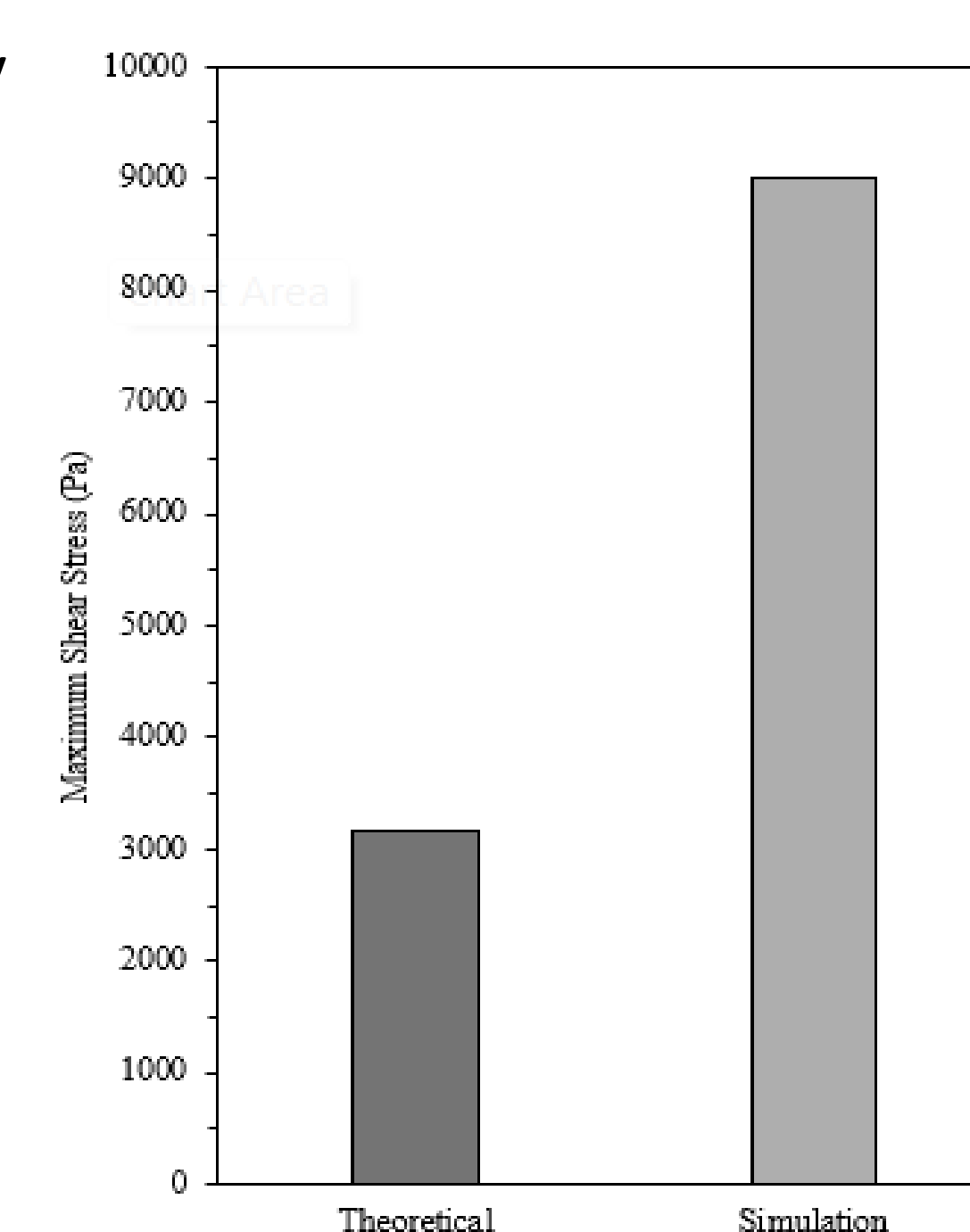


Fig. 7. Maximum Shear Stress comparison between theoretical hand calculations and SolidWorks simulation. Shear stress was calculated on the wrist pin.

Table 1. Parameter comparison of Ford Coyote, Subaru EJ-25, and our model boxer engines.

	Piston Count	Bore	Stroke	Compression Ratio	Ideal Power (HP) at 5500 RPM	Torque
Ford Coyote	8	92.2	92.7	12	463.49	442.58
Subaru EJ-25	4	99.5	79	10	223.85	213.75
Our Boxer	4	24.9	15.8	10	1.79	1.7100

CONCLUSIONS AND FUTURE WORK

- The model EJ25 was successful in maintaining a high RPM
- Crankshaft printed in parts proved to be efficient and effective
- Squared pins resisted rotation and withstood torsional stress
- Future improvement: dried super glue between the connecting rod and crankshaft to reduce friction and prevent fusion